

CRAMER-LAW
LOWER BOUND

SOME EXTENSIONS OF CRLB:

- IF WE ARE INTERESTED IN $\tau(\theta)$, ^{A FUNCTION OF THE PARAMETER θ} SOME FUNCTION τ , THEN THE LOWER BOUND FOR $V(T)$, ^{VARIANCE OF THE ESTIMATOR T OF τ} T UNBIASED FOR $\tau(\theta)$, BECOMES

$$V(T) \geq \frac{\frac{\partial \tau(\theta)}{\partial \theta}}{I(\theta)}$$

FISHER INFORMATION

- **MULTI PARAMETRIC CASE:** $\underline{\theta} = (\theta_1, \theta_2, \dots, \theta_k)$, $k \geq 1$ THEN

FOR T UNBIASED FOR $\underline{\theta}$, ^{PARAM VECTOR} WE HAVE

$$V(T) - I(\underline{\theta})^{-1} \geq 0, \text{ WHERE NOW}$$

$I(\underline{\theta})$ IS THE **FISHER INFORMATION MATRIX**, WITH

$$I(\underline{\theta})_{ij} = E \left[- \frac{\partial^2}{\partial \theta_i \partial \theta_j} \ln f(x; \underline{\theta}) \right]$$

HOW TO FIND UNIQUE ESTIMATIONS

WHAT WE HAVE SEEN, NOT ENOUGH TO FIND UNIQUE, SINCE

- REGULARITY CONDITIONS SOMETIMES NOT VERIFIED
- LOWER BOUND NOT ALWAYS REACHED

EX: $X \sim N(\mu, \sigma^2)$, $\underline{\theta} = (\mu, \sigma^2)$ ^{BOTH PARAMETER UNKNOWN}

$$l(\underline{\theta}; x) = -\frac{m}{2} \ln(2\pi\sigma^2) - \frac{1}{2} \sum_{i=1}^m \frac{(x_i - \mu)^2}{\sigma^2}$$

$$\frac{\partial}{\partial \sigma^2} l(\underline{\theta}; x) = -\frac{m}{2\sigma^2} - \frac{1}{2\sigma^4} \sum_{i=1}^m (x_i - \mu)^2$$

$$\frac{\partial^2}{\partial \sigma^4} \ell(\theta; \underline{x}) = \frac{m}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^m (x_i - \mu)^2$$

$$\begin{aligned} I(\sigma^2) &= \mathbb{E} \left[-\frac{\partial^2}{\partial (\sigma^2)^2} \ell(\theta; \underline{x}) \right] = -\frac{m}{2\sigma^4} + \frac{1}{\sigma^6} \sum_{i=1}^m \mathbb{E}(x_i - \mu)^2 \\ &= -\frac{m}{2\sigma^4} + \frac{1}{\sigma^6} \cdot \sum_{i=1}^m \sigma^2 \\ &= -\frac{m}{2\sigma^4} + \frac{m}{\sigma^4} = \frac{m}{2\sigma^4} \end{aligned}$$

VARIANCE

OPPOSITE

By CRLB: $V(T) \geq \frac{2\sigma^4}{m}$, T ANY UNBIASED ESTIMATOR OF σ^2 .

SAMPLE VARIANCE

WE KNOW THAT S^2 UNBIASED BUT ITS VARIANCE IS

$$V(S^2) = \frac{2\sigma^4}{m-1} > \frac{2\sigma^4}{m} \quad \text{LOWER BOUND NOT REACHED}$$

L

$\Rightarrow$ WE NEED NONE INSTRUMENTS THAN CRLB TO FIND UNVUS

RAO-BLACKWELL THEOREM

SOME FUNCTION "TAU" OF θ

W UNBIASED ESTIMATION OF $\tau(\theta)$, AND T SUFFICIENT ESTIMATION FOR θ .

RAO-BLACKWELL

RB THEOREM TELLS THAT $W^* = \mathbb{E}[W|T]$ IS AGAIN UNBIASED FOR $\tau(\theta)$

BUT IT IS ALSO UNIFORMLY BETTER THAN W :

$$\begin{cases} \mathbb{E}(W^*) = \tau(\theta) \\ V(W^*) \leq V(W) \end{cases}$$

WE CAN IMPROVE AN ESTIMATOR BY GOING FROM ONE ESTIMATOR TO ITS EXPECTED VALUE GIVEN A SUFFICIENT STATISTIC

IN WORDS: IT IS POSSIBLE TO IMPROVE AN UNBIASED ESTIMATION

BY CONDITIONING TO A SUFFICIENT STATISTIC (RAO-BLACKWELLIZATION)

RAO-BLACKWELLIZATION

EX:

WANT TO ESTIMATE p

JUST LOOK VERY FIRST X AND IGNORE THE REST OF THE DATA

$$X \sim \text{Bern}(p)$$

$$W = X_1$$

W UNBIASED SINCE $\mathbb{E}(X_1) = p$

$$\text{Also: } V(X_1) = p(1-p) \quad \text{--- VARIANCE OF BERNULLI VARIABLE}$$

WE KNOW A SUFF. STATISTIC FOR p IS $T = \sum_{i=1}^m X_i \sim \text{Bin}(m, p)$

RAO-BLACKWELL VERSION OF THE ESTIMATOR

WE NEED THE CONDITIONAL

BY DEFINITION

W CAN BE EITHER 0 OR 1 (IS BERNULLI)

$$P(W=w | T=t) = \frac{P(W=w, T=t)}{P(T=t)}$$

$$W \in \{0, 1\}$$

Bim(m-1, p)

$$= \frac{P(W=w, T-W=t-w)}{P(T=t)} = \frac{P(W=w) P(T-W=t-w | W=w)}{P(T=t)}$$

Bim(1, p)

Bim(m, p)

$$= \frac{p^w (1-p)^{1-w} \cdot \binom{m-1}{t-w} p^{t-w} (1-p)^{(m-1)-(t-w)}}{\binom{m}{t} p^t (1-p)^{m-t}}$$

$$= \frac{\binom{m-1}{t-w}}{\binom{m}{t}} = \frac{(m-1)!}{(t-w)!(m-1-t+w)!} \cdot \frac{t!(m-t)!}{m!}$$

TWO CASES

$$\begin{aligned} & \text{For } w=0: \frac{(m-1)!}{t!(m-1-t)!} \cdot \frac{t!(m-t)!}{m!} = \frac{m-t}{m} = 1 - \frac{t}{m} \\ & \text{For } w=1: \frac{(m-1)!}{(t-1)!(m-t)!} \cdot \frac{t!(m-t)!}{m!} = \frac{t}{m} \end{aligned}$$

PROB OF SUCCESS

$$\Rightarrow W|T=t \sim \text{Berm}\left(\frac{t}{m}\right) \Rightarrow W^* = E[W|T] = T/m = \frac{\sum X_i}{m} = \bar{X}$$

$$E(\bar{X}) = E(X_1) = p \quad \text{AND} \quad V(\bar{X}) = \frac{p(1-p)}{m} < V(X_1) = p(1-p)$$

FINALLY, TO REACH AN UNBIASED T OF $\tau(\theta)$ THAT IS ALSO UMVUE
DO RAO-BLACKWELL $\checkmark$ A SUFFICIENT AND COMPLETE STATISTIC

IN OTHER WORDS: W UNBIASED $\Rightarrow W^* = E[W|T]$ AND T SUFFICIENT + COMPLETE
 $\Rightarrow W^*$ IS UMVUE (LEHMAN-SHEFFER THEOREM) MINIMAL SUFFICIENT STATISTIC

UNIFORMLY MINIMUM VARIANCE UNBIASED ESTIMATOR

CONSISTENCY

SOMETIMES ESTIMATORS ARE EVALUATED ACCORDING TO PROPERTIES THAT ARE VALID "ASYMPTOTICALLY", THAT IS BY PRETENDING THAT $M \approx +\infty$, IN CONTRAST WITH FINITE-SAMPLE PROPERTIES.

NOT ANY MORE ONE ESTIMATION, BUT A SEQUENCE OF ESTIMATIONS

$$\{T_m = t_m(x_1, x_2, \dots, x_m)\}, \quad m \rightarrow +\infty$$

1) THE SEQUENCE $\{T_m\}$ IS (WEAK) CONSISTENT FOR θ IF $\forall \theta \in \Theta$

$$\lim_{m \rightarrow +\infty} P(|T_m - \theta| < \epsilon) = 1, \quad \text{ANY } \epsilon > 0$$

SMALL EVENTUALLY GO TO 1

IF I INCREASE THE SAMPLE SIZE THE PROB. OF MY ESTIMATOR BEING CLOSER AND CLOSER TO THE TRUE PARAMETER WILL BE HIGHER

(AS WE INCREASE m , T_m CLOSER AND CLOSER TO θ)

AKA: CONSISTENT IN PROBABILITY -

EQUIVALENTLY: $\lim_{m \rightarrow +\infty} P(|T_m - \theta| \geq \epsilon) = 0 \quad \forall \theta \in \Theta, \forall \epsilon > 0$

2) THE SEQUENCE $\{T_m\}$ IS ASYMPTOTICALLY UNBIASED IF $\forall \theta \in \Theta$

$$\lim_{m \rightarrow +\infty} B_{\theta}(T_m) = 0$$

BIAS

WHEN YOU INCREASE YOUR SAMPLE SIZE TO TWO YOUR BIAS IS = 0

↳ IF YOU COLLECT ENOUGH DATA THE DISTORTION DISAPPEARS

(WITH SAMPLE LARGE ENOUGH, NO DISTORTION)

3) $\{T_m\}$ (STRONG) CONSISTENT FOR θ IF $\forall \theta \in \Theta$

$$\lim_{m \rightarrow +\infty} MSE_{\theta}(T_m) = 0$$

WHEN YOU INCREASE THE SAMPLE SIZE TO TWO ALSO THE MEAN SQUARE ERROR GOES TO 0

MEAN SQUARE ERROR

AKA: CONSISTENT IN MSE

NOTE: STRONG ^③ CONSISTENCY IMPLIES WEAK ^① CONSISTENCY

NOTE 2: STRONG CONSISTENCY IS EQUIVALENT, THANKS TO $MSE_{\theta}(T_m) = V(T_m) + B_{\theta}(T_m)^2$

STRONG CONSISTENCY MEANS HAVING A SEQUENCE OF ESTIMATORS WHICH ARE BOTH ASYMPTOTICALLY UNBIASED AND A VARIANCE THAT APPROACHES 0

$$\begin{cases} \lim_{n \rightarrow \infty} B_{\theta}(T_n) = 0 & (\text{ASYMPTOTICALLY UNBIASED}) \\ \lim_{n \rightarrow \infty} V(T_n) = 0 & (\text{VANISHING VARIANCE}) \end{cases}$$

EX:

SAMPLE MEAN

$$X \sim N(\mu, \sigma^2) \quad \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad \bar{X}_n \sim N(\mu, \sigma^2/n)$$

CHECK THE PROB. OF THE DISTANCE BEING LESS THAN ϵ

$$P(|\bar{X}_n - \mu| < \epsilon) = P(-\epsilon < \bar{X}_n - \mu < +\epsilon)$$

$$= P(\bar{X}_n - \mu < +\epsilon) - P(\bar{X}_n - \mu < -\epsilon)$$

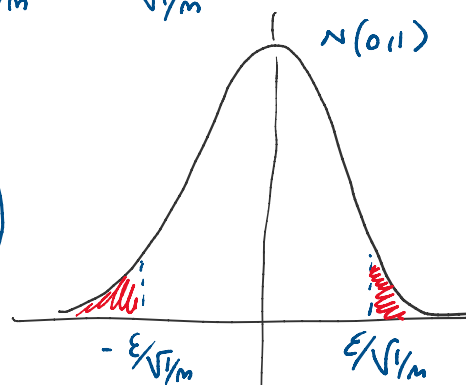
$$= P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} < \frac{\epsilon}{\sigma/\sqrt{n}}\right) - P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} < -\frac{\epsilon}{\sigma/\sqrt{n}}\right)$$

CUMULATIVE DISTRIBUTION FUNCTION
← CDF of $N(0,1)$

$$= \Phi\left(\frac{\epsilon}{\sigma/\sqrt{n}}\right) - \Phi\left(-\frac{\epsilon}{\sigma/\sqrt{n}}\right)$$

$$= \Phi\left(\frac{\epsilon}{\sigma/\sqrt{n}}\right) - (1 - \Phi\left(\frac{\epsilon}{\sigma/\sqrt{n}}\right))$$

$$= 2\Phi\left(\frac{\epsilon}{\sigma/\sqrt{n}}\right) - 1$$



$$= 2\Phi(\epsilon\sqrt{n}) - 1$$

SOONER OR LATER APPROACH 1

$$\xrightarrow{n \rightarrow \infty} 2\Phi(+\infty) - 1 = 2 \cdot 1 - 1 = 1$$

THEN $\bar{X}_n$ WEAKLY CONSISTENT FOR μ .

EX: $X \sim N(\mu, \sigma^2)$. $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. $\theta = (\mu, \sigma^2)$

LAST TIME

$$B_{\theta}(\hat{\sigma}^2) = -\frac{1}{n} \sigma^2 \quad \text{AND} \quad \text{MSE}_{\theta}(\hat{\sigma}^2) = \frac{2n-1}{n^2} \sigma^4$$

$\hat{\sigma}^2$ IS BIASED, BUT ASYMPTOTICALLY UNBIASED SINCE

$$\lim_{n \rightarrow \infty} B_{\theta}(\hat{\sigma}^2) = \lim_{n \rightarrow \infty} -\frac{1}{n} \sigma^2 = 0$$

ALSO $\text{MSE}_{\theta}(\hat{\sigma}^2) \rightarrow 0 \Rightarrow \hat{\sigma}^2$ STRONG CONSISTENT FOR σ^2

L

ASYMPTOTIC PROPERTIES OF MLE

IF $\hat{\theta}$ IS ^{MAXIMUM LIKELIHOOD ESTIMATOR} MLE OF θ THEN (UNDER REGULARITY CONDITIONS):

- $\{\hat{\theta}_m\}$ IS STRONG CONSISTENT FOR θ
- $\{\hat{\theta}_m\}$ IS ASYMPTOTICALLY UNBIASED (BUT WE COULD HAVE A FINITE-SAMPLE BIAS)
- $\{\hat{\theta}_m\}$ IS ASYMPTOTICALLY EFFICIENT, IN THAT ITS VARIANCE APPROACHES THE CRLB:

$$\lim_{m \rightarrow \infty} \frac{V(\hat{\theta}_m)}{1/I_m(\theta)} \rightarrow 1$$

- $\{\hat{\theta}_m\}$ CONVERGES IN DISTRIBUTION TO A NORMAL RANDOM VARIABLE:

$$\lim_{m \rightarrow \infty} P\left(\frac{\hat{\theta}_m - \theta}{\sqrt{1/I_m(\theta)}} \leq z\right) = \Phi(z)$$

EX: $X \sim N(\mu, \sigma^2)$. $\{\hat{\sigma}_m^2\}$ IS STRONGLY CONSISTENT FOR σ^2 (PREVIOUS EX.)

$$V(\hat{\sigma}^2) = \frac{2(m-1)}{m^2} \sigma^4 \quad (\text{LAST TIME})$$

$$\text{CRLB} = \frac{1}{I(\sigma^2)} = \frac{2}{m} \sigma^4 \quad . \quad \text{THEN FROM THE ABOVE PROPERTIES:}$$

$$\begin{aligned} \Rightarrow \lim_{m \rightarrow \infty} \frac{V(\hat{\sigma}^2)}{1/I(\sigma^2)} &= \lim_{m \rightarrow \infty} V(\hat{\sigma}^2) \cdot I(\sigma^2) = \lim_{m \rightarrow \infty} \left(\frac{2(m-1)}{m^2} \sigma^4 \right) \cdot \left(\frac{m}{2} \cdot \frac{1}{\sigma^4} \right) \\ &= \lim_{m \rightarrow \infty} \frac{m-1}{m} = 1 \quad \text{INDEED AS EXPECTED} \end{aligned}$$

$$\text{ALSO: } \frac{\hat{\sigma}^2 - \sigma^2}{\sqrt{2/m \sigma^4}} \approx N(0,1) \quad \text{OR} \quad \hat{\sigma}^2 \approx N\left(\sigma^2, \frac{2}{m} \sigma^4\right)$$

(CAN BE WRITTEN IN TWO WAYS)

PARENTHESIS ON INEQUALITIES

MARKOV INEQUALITY: LET $X \geq 0$ ^{RAND. VARIABLE} RV, $t > 0$ REAL NUMBER.

$$\text{THEN } P(X \geq t) \leq \frac{E(X)}{t}$$

THERE IS AN UPPER BOUND TO THE PROB. THAT X WILL ASSUME "LARGE" VALUES.

EX: X : PRODUCT PRICE WITH $E(X) = \mu$. WHAT ABOUT $P(X \geq \overbrace{10\mu}^t)$?
 $\downarrow$ TO THE THE MEAN OF X

$$P(X \geq 10\mu) \leq \frac{\mu}{10\mu} = 1/10$$

$\Rightarrow$ PROB. OF PRICE ABOVE 10μ IS LESS THAN 10%.

CHEBYSHEV'S INEQUALITY: LET X RV, $y > 0$ POSITIVE REAL. THEN

$$P(|X - E(X)| \geq y) \leq \frac{V(X)}{y^2}$$

THERE IS AN UPPER BOUND TO PROB. OF X DEVIATING FROM ITS MEAN.

EX: $X \sim N(\mu, \sigma^2)$. WHAT ABOUT $P(|X - \mu| > 3\sigma)$?

• USING CHEBYSHEV (IGNORING NORMALITY): $P(|X - \mu| > 3\sigma) \leq \frac{\sigma^2}{(3\sigma)^2} = 1/9 \approx 0.11$

• USING NORMALITY: $P(|X - \mu| > 3\sigma) = P\left(\frac{X - \mu}{\sigma} \leq -3\right) + P\left(\frac{X - \mu}{\sigma} \geq +3\right)$

$$= \Phi(-3) + (1 - \Phi(+3))$$

$$= (1 - \Phi(3)) + (1 - \Phi(+3)) = 2(1 - \Phi(3))$$

$$= 0.0028$$

$$< 0.11$$

$\therefore$ THE NUMBER IS BELOW THE UPPER BOUND PROVIDED BY CHEBYSHEV

AS EXPECTED FROM THE INEQ.

FROM THE ABOVE, THERE CAN BE A LARGE DISTANCE ~~BETWEEN~~ BETWEEN
TRUE PROB. AND THE UPPER BOUND $\Rightarrow$ COST TO PAY FOR A
RESULT THAT IS VALID FOR
ANY DISTRIBUTION OF X

BIG TOPIC

CONFIDENCE INTERVALS

MORE INFO: MARGIN OF ERROR AROUND THE POINT

WE MOVE FROM POINT ESTIMATION TO INTERVAL ESTIMATION

POINT ESTIMATION: SINGLE VALUE
"REASONABLY" ASSIGNED TO θ UNKNOWN PARAMETER
EX: UNKNOWN p
% PEOPLE VOTING FOR A POLITICAL PARTY

INTERVAL ESTIMATION: "RANGE" OF REASONABLE VALUES TO θ
 $\Rightarrow$ WE PROVIDE A QUANTIFICATION OF UNCERTAINTY MORE THAN A SINGLE POINT, YOU INCREASE THE READINESS OF THE OUTPUT
WE TRY TO QUANTIFY THE UNCERTAINTY

THE INTERVAL IS ALWAYS ASSOCIATED TO A "CONFIDENCE LEVEL" (90%, 95%, 99%)

$\hookrightarrow$ HOW CONFIDENT WE CAN BE THAT THE INTERVAL CONTAINS THE TRUE θ

INTERPRETATION : 90% MEANS THAT IF I COLLECT MANY DATASETS,
OF LEVEL
IN 90% OF THEM, THE INTERVAL CONTAINS θ

(STILL, IN OUR UNIQUE DATASET, WE DON'T KNOW IF IT BELONGS TO 90% OR TO 10%.)

CONFID. LEVEL : $1 - \alpha \Rightarrow CI_{1-\alpha}(\theta)$

- Lev 90% $\rightarrow \alpha = 0.1$
- Lev 95% $\rightarrow \alpha = 0.05$
- Lev 99% $\rightarrow \alpha = 0.01$

$\Theta \sim X$:

$X \sim \text{Bern}(p)$... point estimation... $\hat{p} = 0.4$

$$CI_{95\%}(p) = [0.38, 0.42]$$

THE PARAMETER IS WITH 95%
CONFIDENCE BETWEEN 0.38
AND 0.42

$$\text{on } CI_{99\%}(p) = [0.36, 0.44]$$

MORE CONFIDENT THAT $CI \ni p$
SINCE WIDER, BUT AT THE
COST OF LESS PRECISION

WITH 99% WE HAVE A
WIDER RANGE

IF NOTHING ELSE CHANGES
(SAME DATASET, SAME POINT ESTIMATION)
THE ONLY WAY TO BE MORE CONFIDENT
IS TO ENLARGE THE INTERVAL

THE MARGIN OF ERROR AROUND
THE TRUE POINT ESTIMATOR $\hat{p} = 0.4$
DEPENDS ON YOUR CHOICE OF THE
CONFIDENCE LEVEL

$\Rightarrow$ AT A GIVEN $1-\alpha$, SMALLER CIs ARE MORE INFORMATIVE

OTHER DEPENDENCIES:

IF YOU INCREASE N (NUMBER OF DATA) $\Rightarrow$ SMALLER CONFIDENCE INTERVAL CI

• IF YOU INCREASE $N \Rightarrow$ MORE DATA $\Rightarrow$ MORE INFO $\Rightarrow$ SMALLER CI
EXPECTED

• IF YOU INCREASE $1-\alpha$ $\Rightarrow$ MORE CONFIDENCE $\Rightarrow$ WIDER CI

• IF σ^2 HIGHER $\Rightarrow$ MORE UNCERTAINTY $\Rightarrow$ WIDER CI

MORE FORMALLY, $\underline{X} = (X_1, \dots, X_n)$ RANDOM SAMPLE, $X \sim \text{POPULATION}(\theta)$, WE
ESTIMATE θ WITH THE INTERVAL $[L_1, L_2]$ WITH CONF. LEVEL $1-\alpha$

RANDOM VARIABLES

$$P(L_1 \leq \theta \leq L_2) = 1-\alpha$$

LOWER BOUND
OF CI

UPPER BOUND OF CI

$\forall \theta \in \Theta$

$A := L_2 - L_1$ WIDTH OF CI